The Quadratic Formula

For a quadratic equation in standard form

$$ax^2 + bx + c = 0, \qquad a \neq 0,$$

the solutions are

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}.$$

Before substituting

Write the equation in standard form and identify a, b, and c with their signs. Use parentheses when substituting negative values.

- a, b, c are numbers

Use the Quadratic Formula

Solve

$$6x^2 - 5x - 4 = 0.$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

1. Identify a, b, and c.
2. Substitute into the formula without simplifying.
3. Simplify the discriminant and radical.
4. Reduce each solution.

$a=6,\ b=-5,\ c=-4.$

$$x = \frac{5 \pm \sqrt{25 - 4 \cdot 6 \cdot (-4)}}{2 \cdot 6}$$

$$= \frac{5 \pm \sqrt{25 + (-4 \cdot 6 \cdot -4)}}{12} = \frac{5 \pm \sqrt{121}}{12} = \frac{5 \pm 11}{12}$$

$\sqrt{121} = 11$

$5 - 11$

$$= \frac{16}{12} \text{ or } \frac{-6}{12}$$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

The Discriminant

The expression under the radical is the **discriminant**:

$$D = b^2 - 4ac.$$

Value of D	Number of real solutions	Type of solutions
$D > 0$, **perfect square**	two	rational
$D > 0$, **not a perfect square**	two	irrational
$D = 0$	one repeated solution	rational
$D < 0$	no real solutions	two complex conjugates

$\sqrt{-1} = i$

The discriminant predicts the number of x-intercepts before we solve.

- $D = b^2 - 4ac$ is a number.
- example of perfect square: 4, $\sqrt{4} = 2$
- counter example: 3, $\sqrt{3}$
- if $D = 0$, ex: $(x-2)^2 = 0$

  $x - 2 = \pm 0$

  $x = 2 \pm 0$ or $2 - 0$

  $x = 2$
- if $D < 0$

  ex: $x = 1 \pm \sqrt{-2}$

  $= 1 \pm \sqrt{-2} \cdot \sqrt{2}$

  $= 1 \pm i\sqrt{2}$

Classify Before Solving

Find the discriminant. State the number and type of solutions; do not solve the equations.

1.

$$6x^2 - x - 15 = 0$$

$a=6,\ b=-1,\ c=-15.$

$D = b^2 - 4ac$

$= (-1)^2 - 4 \cdot 6 \cdot (-15)$

$= 1 + 360 = 361 = 19^2$

$D > 0,$ D is perfect square.

so 2 real solution, rational.

2.

$$3x^2 - 4x - 5 = 0$$

$a=3,\ b=-4,\ c=-5$

$D = (-4)^2 - 4 \cdot 3 \cdot (-5)$

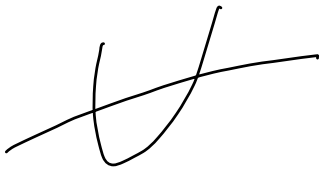
$= 16 + 60 = 76$

so 2 real solution, irrational.

Intersections of Two Functions

To find the intersection points of $y = f(x)$ and $y = g(x)$:

1. Solve $f(x) = g(x)$ for all possible x-values.
2. Substitute each x-value into either original function.
3. Report each intersection as an ordered pair (x, y).



Example. Let $f(x) = x^2 + 5x - 3$ and $g(x) = 2x + 1$. Find all intersection points.

recall:
$f(x) = 3x + 2$
$g(x) = 2x + 4$

$f(x) = g(x)$

$2x + 1 = 2x + 4$

$x = 3$

$f(3) = 3 \cdot 3 + 2 = 10$

$(3, 10)$

Set $f(x) = g(x)$

$x^2 + 5x - 3 = 2x + 1$

$x^2 + 3x - 4 = 0$

$(x+4)(x-2) = 0 \Rightarrow x + 4 = 0$ or $x - 2 = 0$

$x = -4$ or $x = 1$

Plug in $g(x)$,

$g(-4) = 2(-4) + 2$

$= -7$

$g(1) = 2 \cdot 2 + 2 = 3$

$(-4, -7)$ and $(2, 3)$

Functions in Quadratic Form

Some higher-degree equations become quadratic after a substitution.

Substitution pattern

If the equation contains $[u(x)]^2$ and $u(x)$, set

$$v = u(x).$$

Solve the resulting quadratic equation in v, then back-substitute to solve for x.

For $x^4 + x^2 - 12 = 0$, choose a substitution and rewrite the equation in quadratic form.

$x^4 + x^2 - 12 = 0$

set $x^2 = V$, plug V into function.

$(x^2)^2 + x^2 - 12 = 0$

$V^2 + V - 12 = 0$

Solve by Substitution

Solve completely:

$$x^4 + x^2 - 12 = 0.$$

1. Let $v = x^2$.
2. Solve the quadratic equation in v.
3. Back-substitute each value of v.
4. Keep track of real and nonreal solutions.

1) Set $v = x^2$, plug v into the equation.

$x^4 + x^2 - 12 = 0$

$v^2 + v - 12 = 0$

factor: $(v+4)(v-3) = 0$

$v = -4$, $v = 3$

$x^2 = -4$, $x^2 = 3$

$x = \pm\sqrt{-4}$ $\quad$ $x = \pm\sqrt{3}$

$x = \pm i2$

Mixed Quadratic Practice

Choose an efficient method and show complete work.

1. Solve $4x^2 - 8x + 1 = 0$. use Quadratic.
2. Use the discriminant to classify the solutions of $2x^2 + 3x + 7 = 0$.
3. Find the intersections of $y = x^2 - 1$ and $y = x + 1$.
4. Solve $(x+2)^2 + 11(x+2) - 12 = 0$ using substitution.

- factor. ( )( ) = 0
- Quadratic formula. / use some other skill it asks
- x^4, ( )^2, use substitution.

2. $D = b^2 - 4ac$

$= 3^2 - 4 \cdot 2 \cdot 7$

$= -47$

2 complex solution.

3. $x^2 - 1 = x + 1$

$-2 - x \quad -x - 1$

$x^2 - x - 2 = 0$

$(x-2)(x+2) = 0$

$x = 2, \ x = -1.$

$2+1=3$

$(2,3)$

$-2+2=0$

$(-2,0)$

4. $(x+2)^2 + 11(x+2) - 12 = 0$

set $x+2 = V$, plug V into "=":

$V^2 + 11V - 12 = 0$

$(V+12)(V-1) = 0$

$V = -12, \quad V = 1$

$x+2 = -12, \quad x+2 = 1$

$x = -14, \quad x = -1$

1). $2x^2+5x-3=0$ $\quad a=2,\ b=5,\ c=-3$

- find ac.

$$ac = 2 \cdot -3 = -6$$

- find 2 numbers product is $ac=-6$, sum is $b=5$

$6, -2$

$$2x^2+5x-3=0$$

$$2x^2+6x-1x-3=0$$

$$-x-3+\underbrace{2x^2+6x}_{2x}=0$$

$$-1(x+3)+2x(x+3)=0$$

$$(x+3)\cdot(2x-1)=0$$

